Reflective Journal

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In this lab, I learned about the importance of data preparation in machine learning and how raw data needs to be cleaned and organized before it can be used to train a model. Working with the Titanic dataset helped me understand how missing values, categorical data, and numerical features can affect the machine learning process. I learned that data preparation is an important step because the quality of the data can influence how well a model performs.

One of the main concepts I learned was handling missing values through imputation. The Age column in the Titanic dataset contained many missing values, so I used the median to fill them in. I learned that the median is often a better choice than the mean when data contains outliers or is skewed because it is less affected by extremely high or low values. This helped me understand that choosing the right method for handling missing data depends on the type and distribution of the data. I also learned that filling missing values allows us to keep more information instead of deleting rows that contain incomplete data.

Another important concept I learned was one-hot encoding. Machine learning models often need numerical data instead of text, so I used pd.get_dummies() to convert the Sex and Embarked columns into numerical columns. This helped me understand how categorical information can be represented using zeros and ones. I also learned that setting drop_first=True removes one category from each encoded feature to help avoid multicollinearity in certain models.

I also learned about feature scaling and why it is useful for numerical data. I used StandardScaler to scale the Age and Fare columns, so their values would have a mean of approximately zero and a standard deviation of approximately one. This showed me how features with different ranges can be placed on a more comparable scale. However, I learned that scaling is not usually necessary for decision tree models because they make decisions by splitting data based on feature values rather than on the values themselves.

Overall, this lab helped me connect data preparation techniques to the larger machine learning workflow. I now understand that missing values must be handled, categorical features may need to be encoded, and numerical features may need to be scaled depending on the model being used. I also learned that there is not always one correct preprocessing method for every dataset. The best approach depends on the data and the machine learning algorithm. These skills will help me prepare datasets more effectively and understand the steps required before building a machine learning model.